

HL7 ADT Integration Document

Version 1.0

1. Introduction

This document describes the HL7 Version 2.4 ADT (Admission, Discharge, Transfer) message integration between the **CerebralPlus (Source System)** and **Destination System**. The integration allows communication of patient registration, demographic updates, transfers, and discharge events in real-time.

2. Scope

The interface supports the following HL7 ADT event types:

Message Type	Trigger Event	Description
ADT^A01	Register / Admit patient	New patient admission and location assignment
ADT^A02	Transfer	Patient transfer to a new location
ADT^A03	Discharge	Patient discharge or visit closure
ADT^A08	Update patient details	Modification of patient demographics or visit data
ADT^A11	Cancel visit	Reverses a previously sent A01 admission
ADT^A13	Cancel discharge	Reopens a previously discharged visit
ADT^A16 / ADT^A25	Pending discharge / Cancel pending discharge	Flag patient as pending discharge or reverse this status

3. Message Standards

- **HL7 Version:** 2.4
- **Encoding:** | ^ ~ \ &
- **Transport:** MLLP (TCP/IP)
- **Character Set:** UTF-8 (recommended)

Core HL7 segments in use:

- **MSH** – Message Header
- **EVN** – Event Type
- **PID** – Patient Identification
- **PV1** – Patient Visit
- **NK1** – Next of Kin (if applicable)
- **RMI** – Isolation/Precaution Data (A08 only)

4. Sample Messages

4.1 ADT^A01 – Admit / Register Patient

Description:

Creates a new patient visit and assigns the patient to the designated location/ward.

MSH|^~\&|CEREBRALPLUS|PHKL|IWARD|IWARD|20251109112940|ADT^A01^ADT_A01|49361.0
|T|2.4

EVN|A01|20251109112922|||

PID|1||3300055030^^^^MR|ICN^220628085172_D|UENIYUGWAN YIOA ||19220628|F||1301|
^^Others^Lain-Lain^00000^MYS|MYS|0^0|||U|0|||||||MAL|

PV1|1||WWD6^D619^D619|||WWD6^WARD D6 (MEDICAL & SURGICAL)|DYEOPHP^YEOH
POH

HONG|1|||PHKL25IP11000003|15^4^1^C000020027~15^1^99^|||||||||||||2025110911
2919|

NK1|1|||||||||||||||||||||

4.2 ADT^A02 – Transfer Patient

Description:

Transfers a patient to another ward, room, or bed.

MSH|^~\&|CEREBRALPLUS|PHKL|IWARD|IWARD|20250908170900||ADT^A02^ADT_A02|39459.
0|T|2.4

EVN|A02|20250908050840|||

PID|1||2300880685^^^^MR|ICN^801212999998|OESLTLU UEITAU ||19801212|F||1301|No.
1111, Jalan ABC, ABC City^^Kuala Lumpur^Wilayah Persekutuan Kuala
Lumpur^534214^MYS|MYS|0^0123428559|||E|99|||||||MAL|

PV1|1||WWD4^D413-D415^D415|||WWD4^WARD D4 (MEDICAL & SURGICAL)|DASHAP^ASHA
GUPTA|1|||PHKL25IP09000004|15^4^1^C000020027~15^1^99^|||||||||||||2025090811
5443|

4.3 ADT^A03 – Discharge / End Visit

Description:

Indicates the patient has been discharged and the visit is considered closed.

MSH|^~\&|CEREBRALPLUS|PHKL|IWARD|IWARD|20251117012010|ADT^A03^ADT_A03|50692.0|T|2.4

EVN|A03|20251117012004|||

PID|1||3300746940^^^^MR|PP^N7356938|TEST SST PATIENT ||19920409|M||00|Test^^Ayer Hitam^Johor^N/A^MYS|MYS|0^06128764|||0|99|||||||||IND|

PV1|1||WWC7^C706^C706|||WWC7^WARD C7 (EXECUTIVE WARD)|||||||PHKL25IP11000009|15^4^1^C000020027~15^1^99^|||||||||||||20251116210722|||||

4.4 ADT^A08 – Update Patient Details

Description:

Updates patient demographics, contact details, diet, or isolation status.

MSH|^~\&|CEREBRALPLUS|PHKL|IWARD|IWARD|20251118000359|ADT^A08^ADT_A08|50762.0|T|2.4

EVN|A08|20250222120333|||

PID|1||3300269331^^^^MR|ICN^8012129999998|OESLTLU UEITAU ||19480707|F||300|No. 1111, Jalan ABC, ABC City^^Kuala Lumpur^Wilayah Persekutuan Kuala Lumpur^534214^MYS|MYS|0^0|||E|2|||||||||MAL|

PV1|1|O|^|^||DLAMKSP^LAM KAI SENG ||||||||| 15^4^1^C000020027~15^1^99^|||||||||||||DMD, REGD^DIABETIC DIET, REGULAR DIET|||||

NK1|1|^NANGH|40^Daughter|No. 1111, Jalan ABC, ABC City^^Kuala Lumpur^Wilayah Persekutuan Kuala Lumpur^534214^MYS|6012111111253|||||||||||||111103149999

RMI|0|||CI^Contact Isolation, DI^Droplet Isolation|||||||||||||

4.8 ADT^A25 – Cancel Pending Discharge

Description:

Reverses a pending discharge previously flagged by an A16.

MSH|^~\&|CEREBRALPLUS|PHKL|IWARD|IWARD|20250911110055|ADT^A25^ADT_A25|39924.0
|T|2.4

EVN|A25|20250911110051|||

PID|1||3300746898^^^^MR|ICN^800204088828|SL TEST DBT ||19800204|F||200|NO 8, JALAN
MAJUJAYA^^Putrajaya^Wilayah Persekutuan

Putrajaya^80000^MYS|MYS|0^0182882882|||B|99|||||||MAL|

PV1|1||WWC7^C711^C711|||WWC7^WARD C7 (EXECUTIVE WARD)|DALEXLHR^ALEX LEOW
HWONG

RUEY|||||||PHKL25IP07000001|15^4^1^C000020027~15^1^99^|||||||20250702195
704|||||||

5. Transport Protocol

- TCP/IP using MLLP (Minimal Lower Layer Protocol)
- Configurable port per environment
- Supports real-time message delivery

6. Acknowledgements

The receiving system must return:

ACK Code Meaning

AA	Message accepted successfully
AE	Application error (business rule failure)
AR	Message rejected (structural/HL7 error)